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Module 1.2
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CSD 380

The Technology Value Stream

What is it?

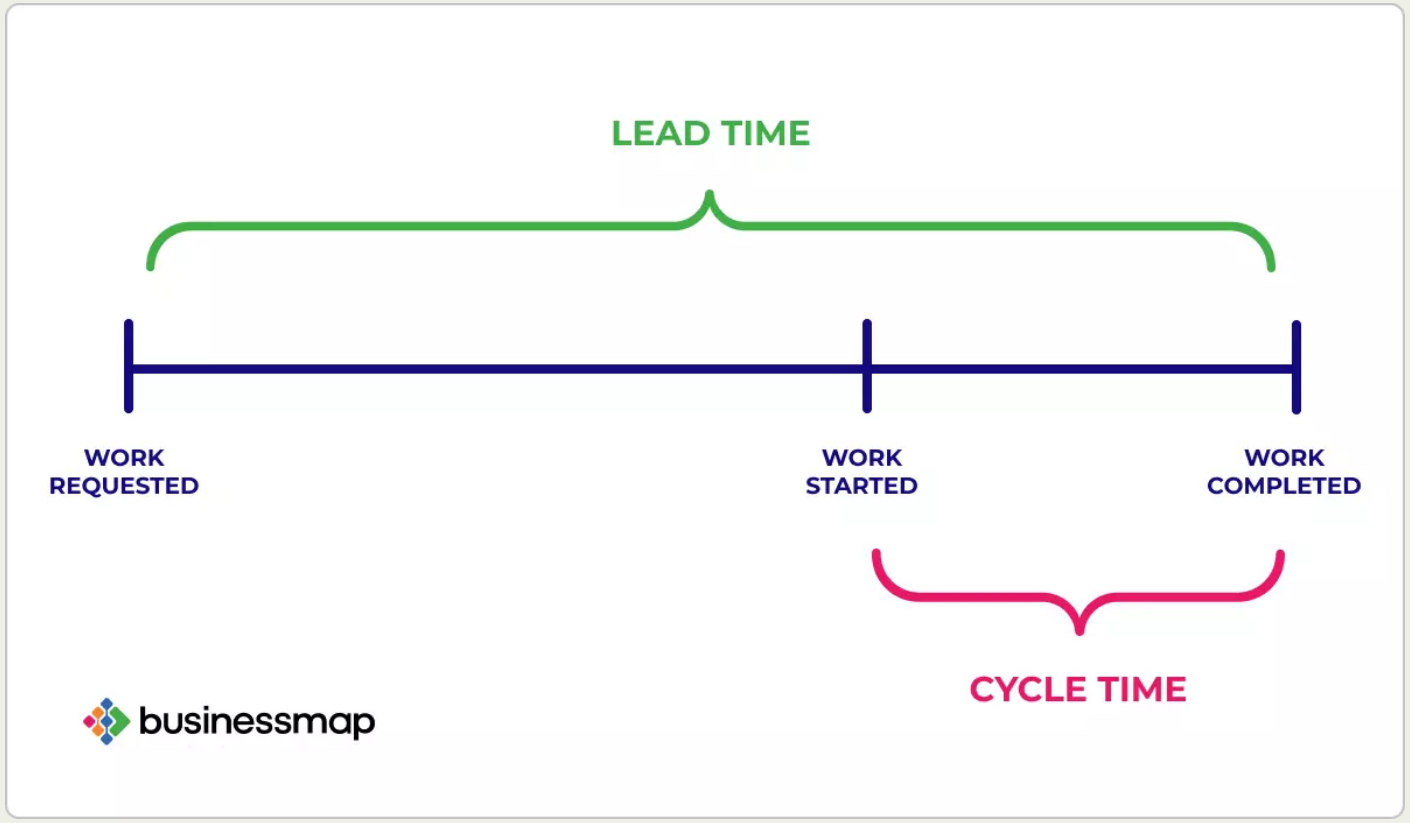
A Technology Value Stream is the process that turns a business idea into something that customers can use.

Business Idea → Development → Testing → Production → Customer

Lead Time VS Processing Time

Lead time is the total time from when a request is made until it is finished which includes time spent waiting.

Processing time is the time spent actively working on the request.

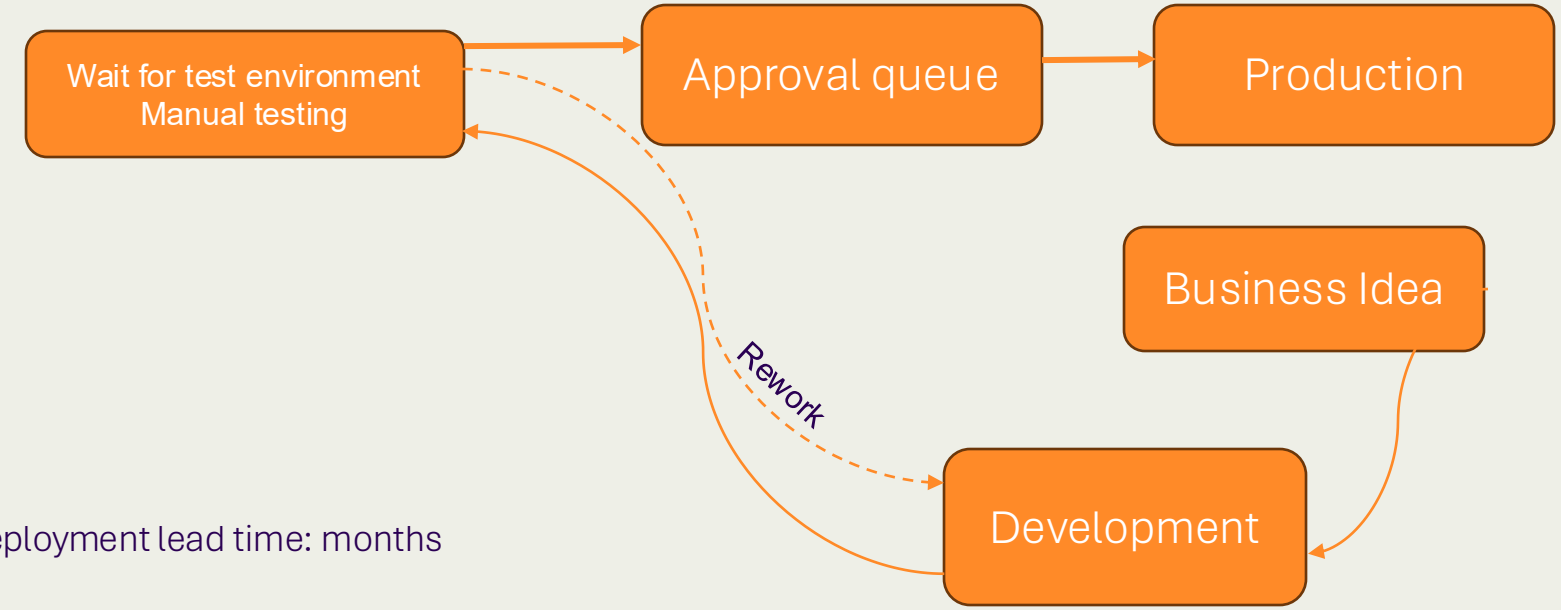


The Common Scenario: Deployments Take Months

- Software systems are closely related, so when one part changes, it can affect other parts.
- Teams have limited test environments and wait for them to become available.
- Testing is often manual, and deployments need approvals.
- Problems may not appear until changes are integrated into the process.

The result is that teams spend days or weeks fixing issues while customers wait for updates.

Why Workflow gets Stuck



Deployment lead time: months

The DevOps Ideal: Deployment in Minutes

- Developers make and check in small changes frequently.
- Automated tests provide quick feedback, enabling teams to find and fix problems sooner.
- The goal is to make a change from version control to production in minutes or, at worst, hours.

Why Shorter Lead Time Matters

- Customers get useful changes earlier.
- Teams get feedback sooner and can correct problems faster.
- Smaller changes are easier to test and troubleshoot, helping limit any disruptions.

Shorter lead time helps teams deliver value faster and respond to problems sooner.

references

Kim, G., Humble, J., Debois, P., Willis, J., & Forsgren, N. (2021). *The DevOps handbook: How to create world-class agility, reliability, & security in technology organizations* (2nd ed.). IT Revolution.

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